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Computational fluid dynamics evaluation of excessive dynamic airway collapse

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A R T I C L E I N F O

Keywords:
Excessive dynamic airway collapse
Computational fluid dynamics
Respiratory functional imaging
Simulation

A B S T R A C T

Background: Excessive dynamic airway collapse, which is often caused by the collapse of the posterior membrane wall during exhalation, is often misdiagnosed with other diseases; stents can provide support for the collapsing airways. The standard pulmonary function tests do not necessarily show change in functional breathing condition for evaluation of these type of diseases.
Methods: Flow characteristics through a patient's airways with excessive dynamic airway collapse have been numerically investigated. A stent was placed to support the collapsing airway and to improve breathing conditions. Computed tomography images of the patient's pre- and post-stenting were used for generating 3-Dimensional models of the airways, and were imported into a computational fluid dynamics software for simulation of realistic air flow behavior. Unsteady simulations of the inspiratory phase and expiratory phase were performed with patient-specific boundary conditions for pre- and post-intervention cases to investigate the effect of stent placement on flow characteristic and possible improvements.
Findings: Results of post-stent condition show reduced pressure, velocity magnitude and wall shear stress during expiration. The variation in wall shear stress, velocity magnitude and pressure drop is negligible during inspiration.
Interpretation: Although Spirometry tests do not show significant improvements, computational fluid dynamics results show significant improvements in pre- and post-treatment results, suggesting improvement in breathing condition.

1. Introduction

Expiratory central airway collapse (ECAC) is the narrowing of the airways during expiration, which is divided into Tracheomalacia (TM)/ Tracheobronchomalacia (TBM) and Excessive Dynamic Airway collapse (EDAC) (Murgu and Colt, 2013). Current methods of assessment of these diseases are challenging due to the similarity of symptoms to other diseases and conventional static medical imaging. Although dynamic imaging has proven to be effective in the diagnosis of these diseases, understanding air flow function and behavior is of great interest to researchers and clinicians for assessment and diagnosis.

Infections, inflammatory disorders, malignancy and extrinsic compression from adjacent structures can result in tracheal disorders; stenosis or an upper airway obstruction, is the manifestation of these diseases. Compared to lower airway obstructions, tracheal obstructions can be life threatening, since there are no collateral ventilations that can distribute the air to other airway branches (Al-Qadi et al., 2013).

The incidence of EDAC and TBM rely heavily on the narrowing airway lumen percentage criteria, historically by more than 50% and recently by more than 80% (Murgu and Colt, 2013) in lumen reduction, and it is reported to be between 4 and 22% of patients with chronic obstructive pulmonary disease (COPD) and/or asthma (Bruno and Mariotta, 2013; Dal Negro et al., 2013; Murgu and Colt, 2013).

Although EDAC is often mentioned interchangeably and incidentally with TBM, there are morphological and pathophysiological differences between these diseases. TM and TBM are characterized by weakness of the cartilaginous structure in the trachea and can extend to one or more primary bronchi, respectively; whereas EDAC is the collapse of the posterior membrane wall during exhalation or coughing, and it is unrelated to cartilage function and collapse (Al-Qadi et al., 2013; Murgu and Colt, 2013). TBM and EDAC can be classified into five domains: functional impairment, morphology, origin, length and

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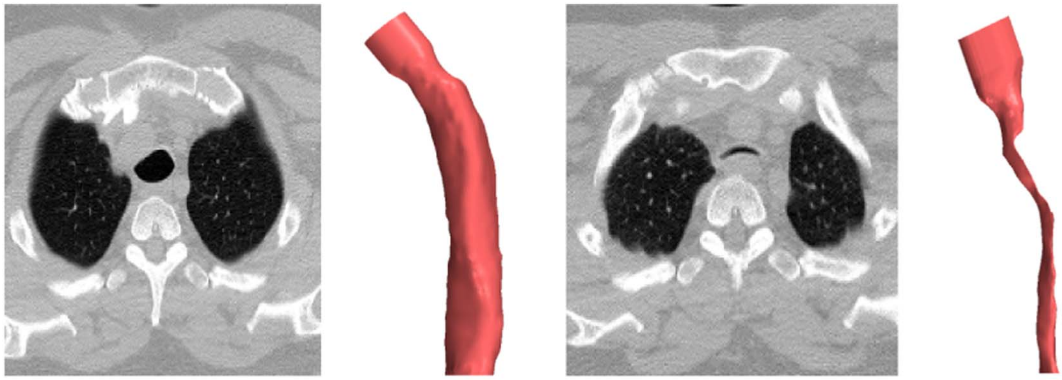

Fig. 1. CT-Images before stent placement at peaks of inspiration (left) and expiration (right) cycles, and their respective 3-D segment models of the trachea.

Table 1
Pulmonary function test (PFT) results of a patient diagnosed with EDAC before and after stenting. Although forced expiratory volume 1(FEV1) divided by forced vital capacity (FVC) does not indicate an obstruction (less than 70%) before intervention, the FEV1 increases (by more than 200 ml), as well as an increase in FEV1/FVC ratio is observed in post interventions. However, these values are neither significant nor sensitive to ECAC diseases.

Pre-stent spirometry		Ref	Pre	%Ref	Post-stent spirometry	Ref	Pre	%Ref
FVC	Liters	4.85	4.31	89	FVC	4.82	4.23	88
FEV1	Liters	3.76	3.10	82	FEV1	3.73	3.39	91
FEV1/FVC	%	78	72	93	FEV1/FVC	77	80	104

location abnormality, and severity of airway collapse (Dal Negro et al., 2013; Murgu and Colt, 2013). However, additional parameters, such as flow behavior, can significantly help with the assessment, pre-operative planning and post-therapy monitoring of these diseases.

Various diseases can be associated to ECAC. Physiological studies demonstrate that collapsed airways in EDAC patients might be the consequence of COPD, resulting from decreased elastic recoil, small airway inflammation and atrophy of elastic fiber, asthma, bronchiectasis and bronchiolitis due to small airway inflammation, and obesity due to positive pleural pressure (Murgu and Colt, 2013).

Different diagnosis modalities can be used for the diagnosis of TBM/EDAC, and can be categorized into invasive and non-invasive diagnosis. The gold standard for dynamic tracheal collapse visualization is by bronchoscopy, which is an invasive procedure. CT scans can be used as a non-invasive tool for diagnosis of ECAC; CT-scans provide additional information (on the vasculature, masses/tumors or parenchymal changes near the trachea) to understand the causes of ECAC (Murgu and Colt, 2013). However, the static inspiratory or expiratory phase cannot truly indicate the degree of airway collapse and thus making dynamic CT-scans a superior choice. Generally, CT-scans are used for pre- and post-operation (e.g. stent) assessment, to monitor possible complications such as mucus obstruction or choke point migration (Murgu and Colt, 2013).

Stents can provide support for the collapsing airways; metallic and silicon stents are used for these types of diseases, with the former being the primary choice due to fewer complications (Dal Negro et al., 2013; Murgu and Colt, 2013). As a foreign body, stents generally may cause disturbances in the clearance of mucosa, resulting in plugging with a tendency to initiate local infection sites (Zarogoulidis et al., 2015). In order to monitor the stent's possible adverse effects, a follow-up bronchoscopy or dynamic CT have been used (Murgu and Colt, 2013). However, the effect of location of stent placement on airflow has not systematically been considered in the pre-clinical setting.

For the past decade, there has been a large body of research covering Computational Fluid Dynamics (CFD) simulations of the airways, with small relevant sample discussed here. There have also been investigators who have focused on CFD and fluid-structural interactions (FSI) of airway obstructions, tracheal stenosis, and stent placements (Malvè et al., 2011a; Malvè et al., 2011b; Pirnar et al., 2015; Zhao et al., 2013). However, to our knowledge, there has not been a CFD investigation for airflow evaluation of EDAC and the effect of stent placement on its prognosis.

FSI simulation could be used to demonstrate the effects of the interactions of air with cartilaginous and muscle structure, but there are major challenges with this approach. The most prominent challenge is the patient specific local structural (trachea) property changes due to various types of diseases. The lack of accurate structural property of this disease makes the traditional FSI impractical.

Also, the majority of CT-based CFD and FSI simulations of the airways only consider the CT-based models of the inspiratory phase. However, in EDAC disease the reduction in the airway lumen and collapse of the trachea are during expiration phase; thus, making the simulations of only inspiratory phase models inaccurate. Here we use four sets of CT-Images. In order to understand the effect of stent placement on flow characteristics, inspiratory and expiratory CT images were used (Fig. 1) in both pre- and post-stenting for 3-Dimensional(3-D) model generation, as well as defining patient specific boundary conditions.

2. Three-dimensional models

CT-images provided by the Long Beach Veterans Administration (LBVA) Healthcare System were transferred to Mimics (Materialise) software for image segmentation and 3-D model generation. The models were then transferred to the STAR-CCM + (Siemens) software for CFD analyses. Four sets of CT-images were used in this study; pre- and post-intervention CT-based models of a patient diagnosed with EDAC were simulated during the inspiratory-expiratory cycle.

The branching airways below the trachea were not included in previous studies (Brouns et al., 2006; Chen et al., 2014; Malvè et al., 2011a; Mylavarapu et al., 2013; Zhang and Kleinstreuer, 2011), with few exceptions (Luo et al., 2007; Xi et al., 2008). In this study, branching sections up to 6–8 generations have been included. Although the main concentration in the trachea and upper airway analysis is the flow behavior in these regions, the upstream pressure could be affected if the lower generations are omitted. It is important to note that the larynx and oropharynx regions have shown to influence the flow behavior downstream in a healthy subject (Taherian et al., 2016). However, the high degree and length of collapse in this patient reduce the impact of upper regions in the trachea and lower generations. The authors are currently investigating these matters.

3. Computational fluid dynamics methods

The effect of EDAC on flow behavior in the trachea has been

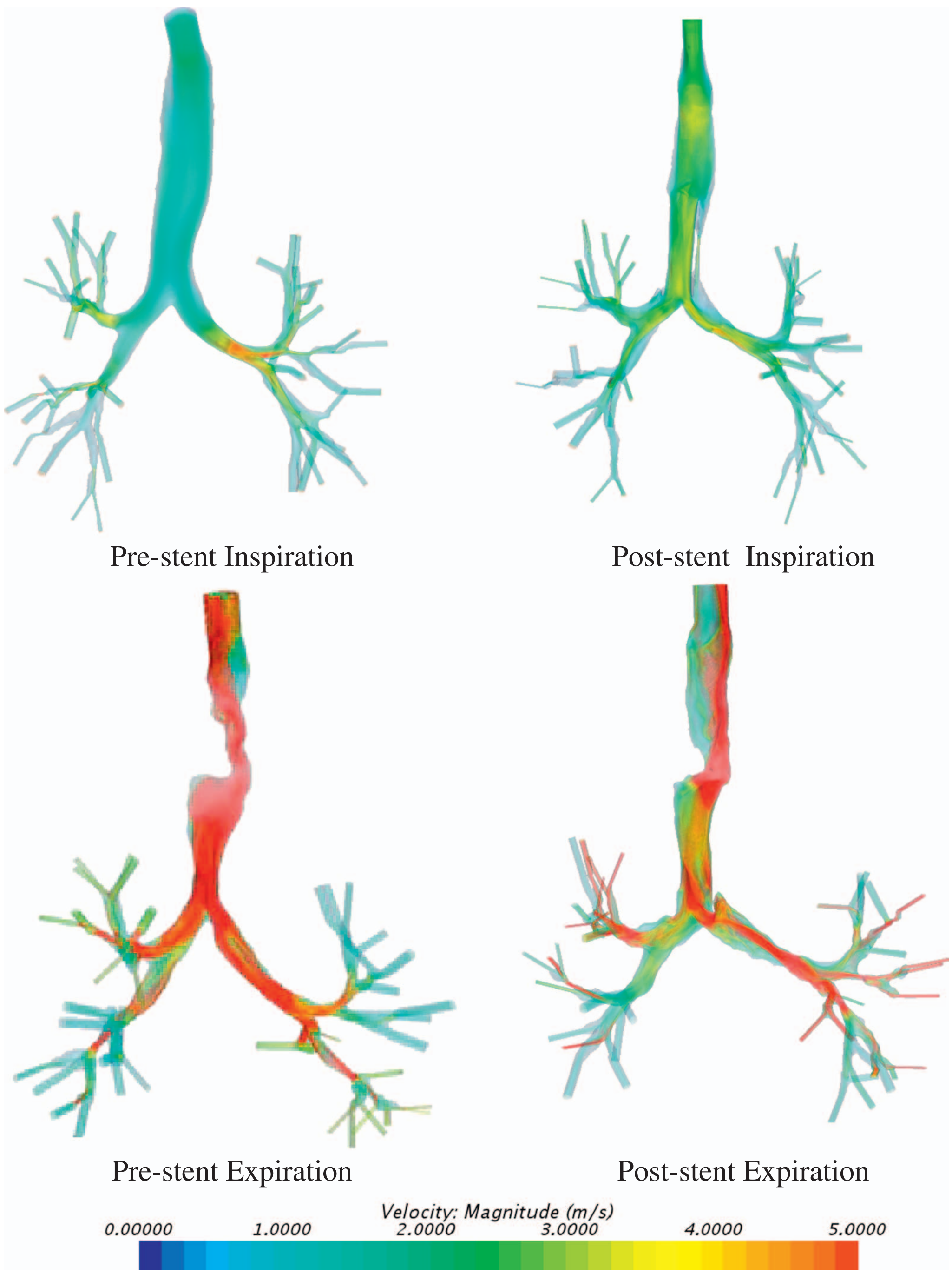

Fig. 2. Velocity magnitude (m/s) pre-stent placement (left) and post-stent placement (right) at peak flow during inspiration (top) and expiration (bottom) cycles. The scale is adjusted for better representation of the results.

investigated. The models include the trachea and lower generation of airways. In comparing the results before and after interventions and across four models, it was essential that boundary conditions were the same for inspiratory and expiratory models, especially at the outlets.

Although some investigators have focused on multidimensional numerical models, including a 3-D CFD coupled with 0-D (Oakes et al., 2013; Wongviriyawong et al., 2012) or 1-D (Lin et al., 2009; Ma and Lutchen, 2006; Ma and Lutchen, 2008; Tawhai, 2004; Tawhai et al., 2000; Tawhai and Lin, 2010) models to simulate the deeper generations of the lung coupled with the larger 3-D airways, most have used constant velocity or generalized zero pressure boundary conditions that might not represent the airflow distribution to different lung lobes. While zero pressure outlets are not realistic, in the absence of appropriate data, these boundary conditions might be suitable in the normal trachea (Calmet et al., 2016; Yousefi et al., 2017), especially when lower regions are not included; however, it might not properly characterize the patient with an obstruction which causes a considerable pressure drop in the upper airways.

We have used a two-step boundary conditions developed in our previous investigations for tracheal stenosis and particle deposition in a human respiratory system (Taherian et al., 2014; Taherian et al., 2015). In this method, pressures assigned at each outlet were according to the patient specific lobar volume changes obtained from the inspiratory-expiratory CT images. Mass flow outlet boundary conditions were assigned at each outlet during the first step to measure the pressure imposed by the lobar volume changes on the branching airways. The results from previous step were then assigned as individual pressure functions at each pressure outlet to make the flow pressure-driven in the second step, and also to increase the stability and better convergence of the simulations. The unsteady simulations of inspiratory

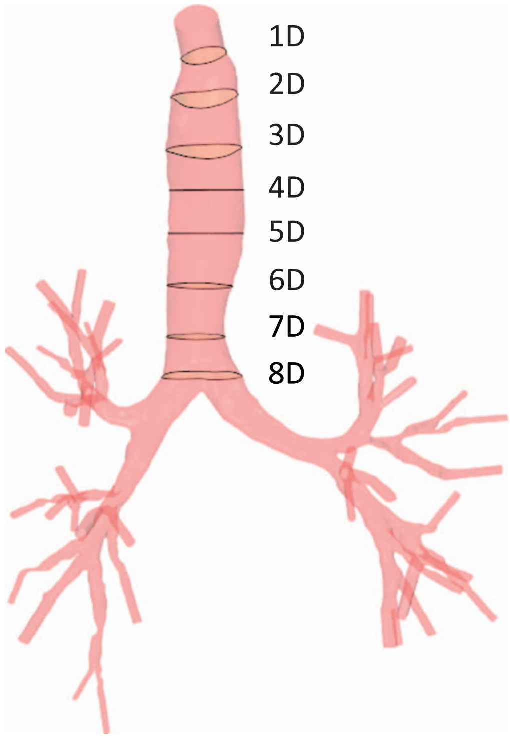

Fig. 3. The location of axial planes in the trachea approximately 1 inlet-diameter (D) distance apart.

and expiratory phases were performed using rigid inspiration and expiration models respectively. Inlet pressure was assumed to be atmospheric pressure with 10% turbulence intensity. The flow governing equations are discretized on the computational domain using second-order finite-volume schemes, and a second-order implicit discretization was employed for the time integration. Unsteady simulations for 2.5 s of inspiration and 2.5 s of expiration were implemented using sinusoidal type flow behavior under resting tidal volume condition.

Extensions were added to inlets and outlets, one and four hydraulic diameters long respectively, to ensure uniformity and continuity. Reynolds-averaged Navier–Stokes (RANS) turbulence modeling and large eddy simulation (LES) are two main approaches for turbulent flow computations in human respiratory system. Although LES can provide instantaneous velocity fluctuations, appropriate RANS models are still widely preferred in turbulent flow computations since the prediction of mean flow behavior is of main interest. Moreover, LES requires 100-fold computing time compared to RANS turbulence models (Sandham, 2005; Versteeg and Malalasekera, 2007; Zhang and Kleinstreuer, 2011). Unsteady RANS, with low Reynolds number K-ω turbulence model, were used for this investigation.

Polyhedral and prism layer meshes were used for all investigations. Three parameters of pressure, velocity, and average WSS were chosen to perform mesh independency analysis and the results for pre-stent, expiratory phase model, were compared. Pressure and velocity were measured at an axial plane in the trachea before the main branching section (8D in Figs. 3–8) and average WSS was measured in the trachea.

The results of 900,000 and 3.2 million cells, when compared with 1.7 million cells demonstrate that pressure changes were 10% and 3%; velocity changes were 2% and 0.6%; and the changes in WSS were 13% and less than 0.5%, respectively. Thus, 1.7 million cells were assumed to be sufficient for these analyses.

4. Results

Spirometry results, one of the pulmonary function test (PFT) measurements, do not necessarily show the percentage of symptom

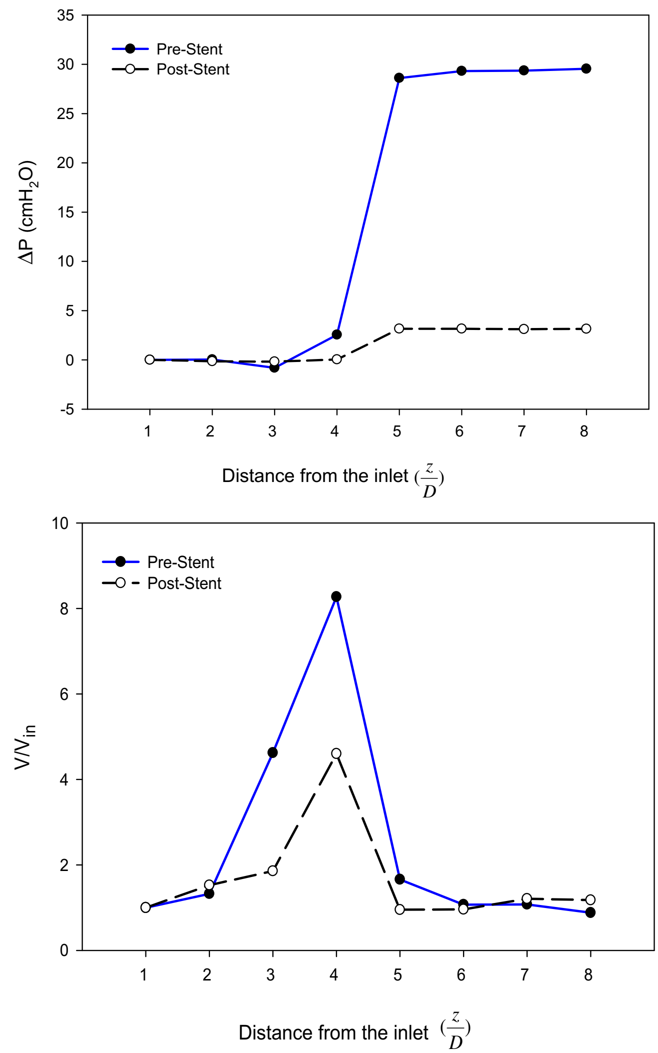

Fig. 4. Pressure (top) and velocity (down) variations at peak flow during expiratory phase. The blue solid lines indicate pre- intervention and dashed black lines indicate post-interventions. ΔP is the local averaged pressure minus the inlet pressure (in cmH_2O), V/Vin is the ratio of local averaged velocity divided by the corresponding inlet velocity, and z is the distance of axial planes in the trachea from the trachea opening downstream divided by the diameter (D) of the inlet. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

improvement before and after stent interventions or tracheoplasty. While PFT might suggest improvements in coughing or pulmonary mechanics, it does not show the degree of airflow improvement measured by forced expiratory volume 1 (FEV_1), which measures the volume of air forcefully exhaled in 1 s (Murgu and Colt, 2013).

In this patient with severe EDAC, the PFTs of pre-and post-stenting (Table 1) show an increase in post stent FEV1 of 290 cm^3, suggesting a decrease in this patient's obstruction and increasing the FEV1/FVC ratio. However, this patient's baseline PFTs was within normal limits confirming PFT's poor sensitivity when evaluating EDAC and/or TBM. The CFD results show flow characteristics, pressure drop, and flow velocity across the trachea, providing functional information which could be used for evaluation of ECAC diseases.

Velocity magnitude variations are shown in Fig. 2. The volume flow rates were assumed to be the same for all cases. The characteristic Reynolds numbers at the entrance of the trachea, upstream of obstruction, at the peak inspiration-expiration cycles were 2040–5400 for pre-stent, and 2350–3350 for post-stent, respectively. The scale bar has been adjusted to clearly show the significant changes in flow characteristic and velocity magnitude during inspiration/expiration for pre-

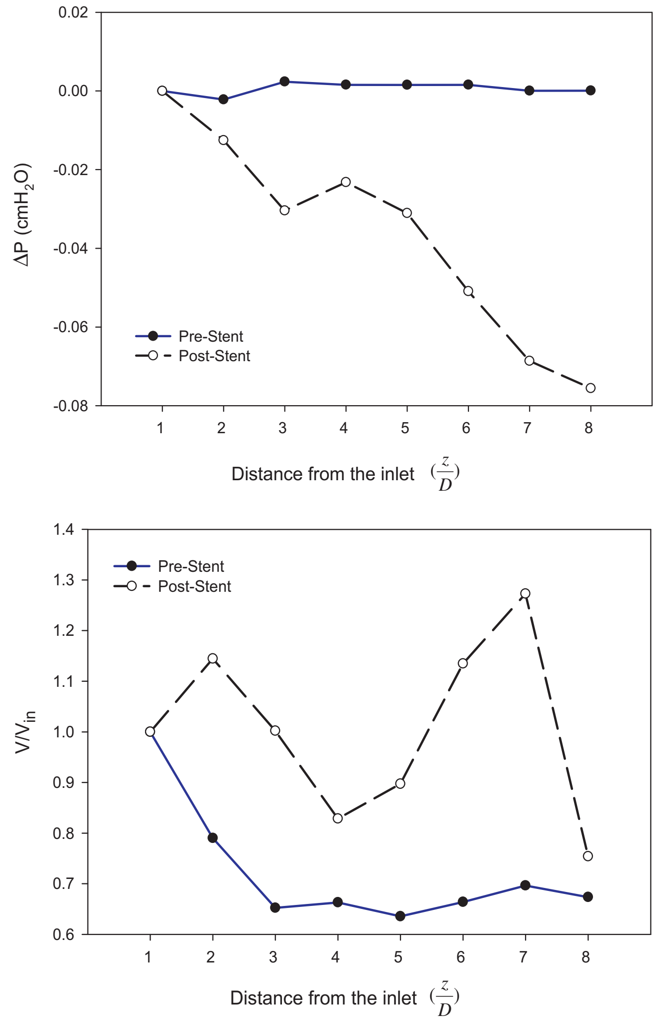

Fig. 5. Pressure (top) and velocity (down) variations at peak flow during inspiratory phase. The blue solid lines indicate pre-intervention and dashed black lines indicate post-interventions. ΔP is the local averaged pressure minus the inlet pressure (in cmH_2O), V/Vin is the ratio of local averaged velocity divided by the corresponding inlet velocity, and z is the distance of axial planes in the trachea from the trachea opening downstream, divided by the diameter (D) of the inlet. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
The averaged wall shear stress (in Pa) variation in the trachea pre- and post-stent placement at peak flow during respiration cycles.

WSS (pa)	Pre-stent	Post-stent
Inspiratory	0.0354	0.0813
Expiratory	3.98	0.793

and post-stent. With pre-stent inspiration, the velocity distribution shows significant low-velocity regions with more air moving to one side of the trachea wall and an uneven increase in mean velocity through the branches. With the stent in place, the mean velocity upstream of the stent is increased and the velocity field is expanded non-uniformly throughout the stent and downstream. There are better mean velocity distributions within the airway branches, in particular through the first airway generation with reduced maximum velocity.

For expiration, the mean velocity contours display significant variations in both pre- and post-stent conditions. Without the stent in place, the mean velocity is increased both upstream and downstream of obstruction with un-even velocity distribution beyond the obstruction. With the stent in place, the mean velocity is reduced upstream and expanded through the trachea. With increased area, the mean velocity is decreased, which indicates improved breathing condition. The flow in branching section has also changed, which may suggest changes in flow distribution to different lung lobes.

Fig. 3 demonstrates the location of the axial planes, which pressure, velocity and turbulence intensity were obtained, in the trachea approximately at one inlet-diameter (D) distance apart.

Fig. 4 shows variations in the mean pressure and velocity in trachea at peak expiratory phase for pre- and post-stenting. The location of the maximum obstruction is at 4D from the inlet.

ΔP is the local averaged pressure difference with respect to the inlet pressure and z/D, where z is the distance of the axial planes in the trachea from the inlet downstream, divided by the diameter (D) of the inlet. Each plane is approximately 1D apart from the next measured plane. For pre-stent, there is a slight decrease in pressure from section 8 to section 4, a significant drop in pressure across the obstruction and then pressure recovery to atmospheric condition. When the stent is in place, there is a slight increase in pressure from section 8 to section 4, and a slight pressure drop across the trachea to atmospheric pressure. The net difference in pressure for pre- and post-stent condition is approximately 26 cmH_2O; which suggests significant improvement in functional breathing.

For the velocity, V/Vin is the ratio of local averaged absolute velocity divided by the corresponding inlet velocity. The normalized mean velocity for pre-stent condition shows an increasing trend from section 8 to section 4, a sharp increase at the location of the obstruction, before it decreases to the inlet value. When the stent is in place, the change in the normalized mean velocity is similar with significantly reduced variations.

Variation in the normalized mean velocity corresponds to the variation in pressure; where, with an increase in pressure, velocity decreases and vice versa. It should be noted that existence of obstructing regions result in areas of high recirculations, both upstream and downstream, which could contribute to velocity gradients and higher wall shear stress, resulting in increased pressure losses.

Fig. 5 shows variation of the mean pressure and normalized velocity for both pre- and post-stent at peak inspiration. For pre-stent condition, pressure slightly decreases initially, then increases approaching the obstruction area and becomes a nearly constant value. The corresponding normalized velocity decreases initially, oscillates slightly across the blockage region. However, when the stent is in place, the pressure drops initially, increases slightly upstream of the stent (stent blockage effect), and then decreases afterward. The normalized velocity initially increases upstream of the stent, declines across the stent, increases again up to $z/D = 7$ and decreases again afterward. In general, according to Bernoulli's principle, velocity is inversely proportional to the pressure. While in most areas, this is true, however for locations immediately upstream of the stent, both pressure and velocity decrease. The decrease in velocity could be due to recirculation areas as a result of stent placement which also contributes to increased wall shear stress and thus additional pressure losses.

Table 2 provides averaged wall shear stress for the entire trachea with and without the stent for both inspiration and expiration. During inspiration, with the stent in place, the averaged shear stress increases which as indicated before could be due to the extensive flow unsteadiness, recirculation and pressure variation. However, for expiration, the averaged shear stress has decreased significantly, by nearly 80%. The decrease in the averaged shear stress agrees with the variations of the pressure and the normalized mean velocity, as shown in Figs. 3 and 4.

Fig. 6 demonstrates contours of the WSS for pre- and post-stenting during expiration. The variation in WSS during inspiration is negligible; while during expiration, the collapsing airway with maximum lumen airway reduction will experience the highest WSS. Although the WSS is significantly reduced in post-intervention at the collapsing region, there is a slight increase in wall shear stress in first branching sections of the

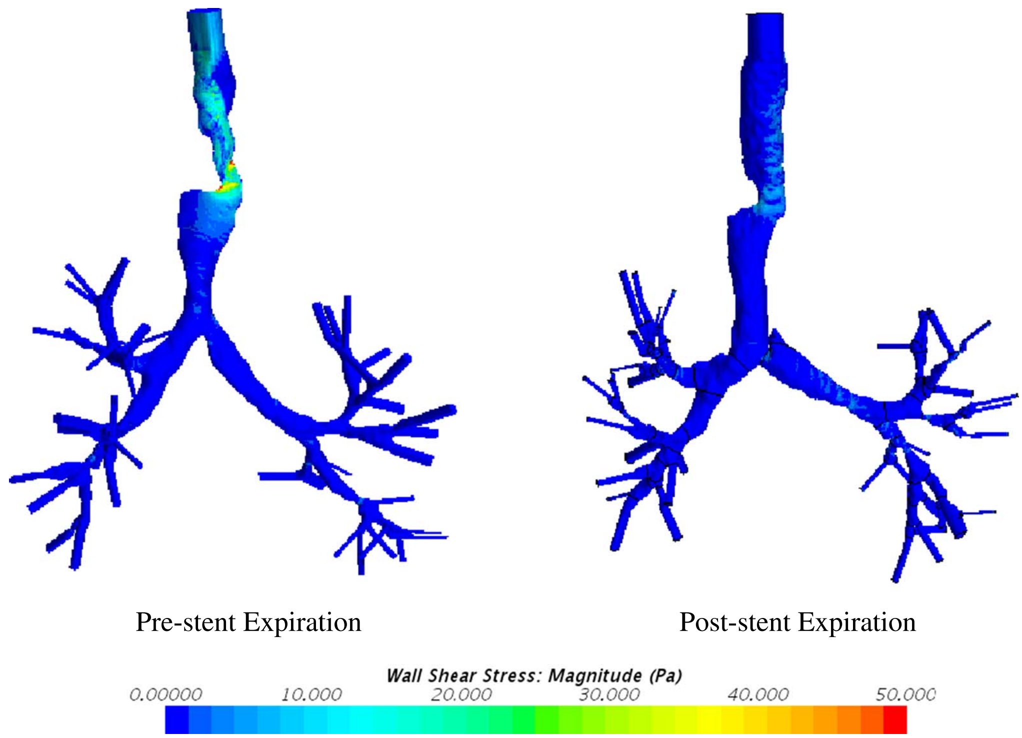

Fig. 6. Wall shear stress variation (in Pa) at peak flow during expiration, pre- and post-intervention.

airways. Variation in shear stress is related to the velocity gradient at the wall. Obstruction, flow recirculation, and branching airways would result in increased WSS. The introduction of the stent removes the blockage and reinstate normal flow movements; however, change in wall compliance, due to stent placement, could create local obstruction with additional stress and increased resistance and turbulence intensities.

Investigation of WSS is important due to the evaluation of stresses experienced by the epithelium lining of the respiratory tract, which acts as a barrier for foreign particulates. High WSS could lead to more mucus secretion and may cause a blockage at the entrance of stent.

Fluctuation in airflow pathing through the airways can be characterized by turbulence intensity (TI). The existence of the stent affects TI both during inspiration and expiration (Fig. 7). For pre-stent inspiration, the increase in TI is seen upstream of the blockage and in some branches were increased velocity is observed. In post-stent inspiration, TI increases across the stent and in the first branches, but not in other branches. The introduction of the stent results in increased velocity with possibly some flow distortions through the stent and the first branches, which has resulted in increased TI.

For pre-stent expiration, the increase in TI is seen in the second-generation branches and after the blockage. Merging of flow from branches results in increased TI and upstream of the obstruction, the jet-like flow acceleration and expansion increases TI.

When the stent is in place, during the expiration, increased TI is observed through branches and beyond the stent; as again increased velocity and expansion throughout the whole region results in increased TI. Fig. 8 shows variations of maximum TI measured on each plane during inspiration-expiration with and without stent in place which are in accordance with the corresponding results in Fig. 7; the stent placement increases TI in both inspiratory and expiratory cases. For patients using inhaled medication, the increased in TI can significantly affect droplet/particle transport and depositions.

Lateral pressure measurements on each side of the obstruction can provide information regarding the degree of obstruction. Pressure differences between theses points with respect to their time series for inspiration and expiration cycles are plotted against each other. Fig. 9, shows pressure-pressure (P-P) curve during quiet breathing. The angle of the resulting curve, when normalized against the maximum local pressure difference, may provide airway's functional information in response to stent placement; change in this angle to 45° could suggest improvements in clinical intervention outcomes. More patients with different degrees of obstruction are needed to increase sensitivity and reproducibility of this test, which is the subject of the authors' ongoing investigation.

5. Discussions

PFTs do not correlate well with the diagnosis or degree of illnesses when dealing with central airway diseases. During exhalation, the overall tracheal volume added to the total volume of air moved is minimal; however, PFT in ECAC might show reduced expiratory flow, notching on the flow-volume (FV) loop, dynamic airway compression (demonstrated as subtracting slow vital capacity from forced vital capacity), a biphasic FV loop or flow oscillations. Still, these findings are neither specific nor sensitive (Murgu and Colt, 2006).

The PFT in EDAC patients can show symptoms of obstruction, but it does not provide any information regarding the severity of the reduced airway lumen. Likewise, it does not show the degree of symptomatic improvement following stent and other correcting interventions. Airflow improvements due to clinical interventions are not indicated by airflow improvement as measurements by FEV1% for EDAC patients; however, interventions outcomes are observed in other factors such as cough reduction, secretion management and improvement in overall pulmonary mechanics (Murgu and Colt, 2013).

During expiration, the increased velocity, averaged wall shear stress and a possible increase in work of breathing, which is the function of pressure and volume, could worsen the condition of the patient by exaggerating the movement of the airway wall in pre-intervention EDAC patients. CFD results of post-intervention show significant improvements in patient's breathing conditions. Although comparison of the results of the pre- and post-intervention show improvements in airflow, Fig. 4 at 4D and Fig. 7 suggest that placement of a longer stent in this patient might have improved the results in keeping the airway

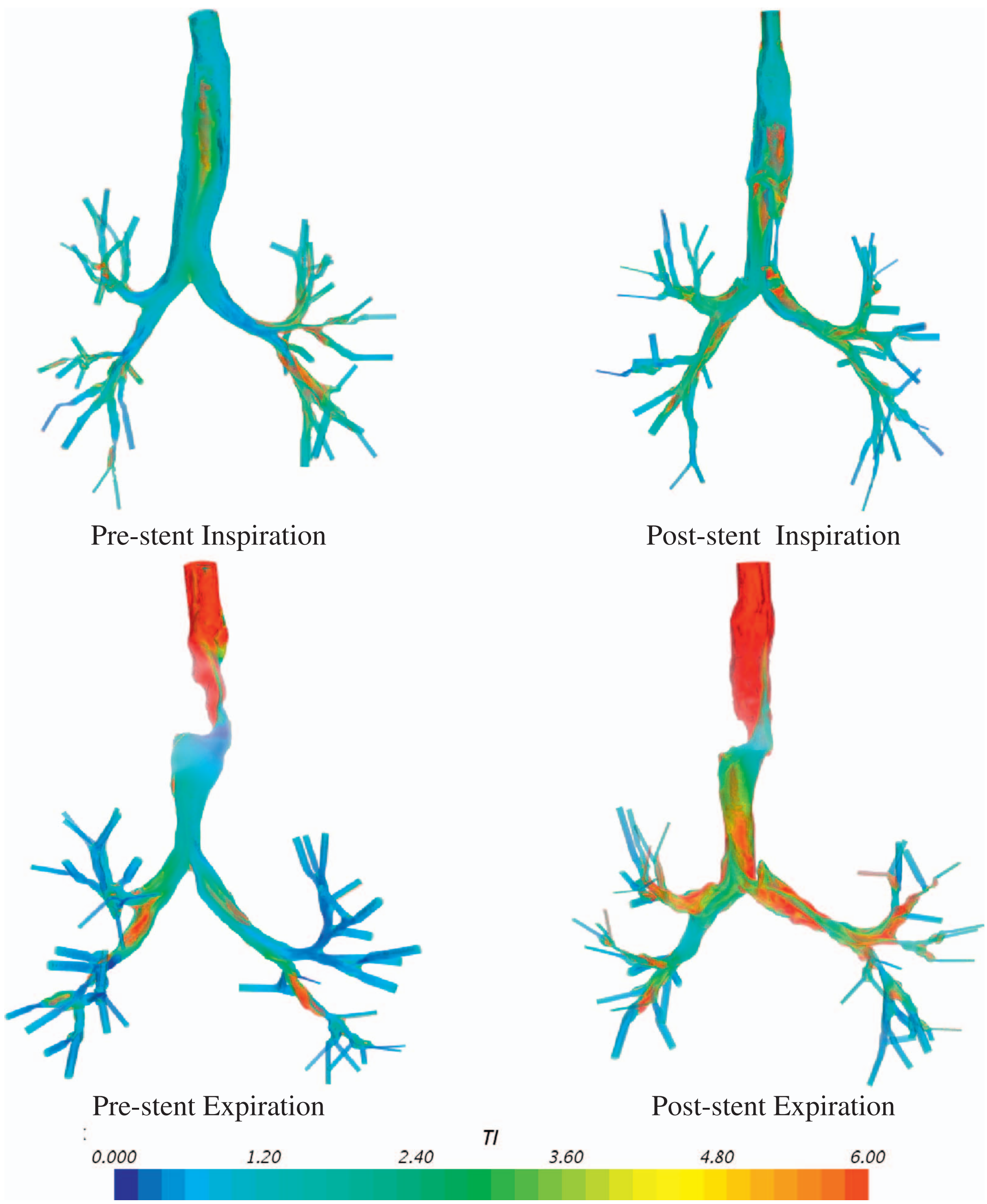

Fig. 7. Turbulent Intensity (TI %), pre-stent placement (left) and post-stent placement (right) at peak flow during inspiration (top) and expiration (bottom) cycles. The scale is adjusted for better representation of the results.

open and reducing velocity variations and WSS.

The similar type of P-P curve flow behavior is observed by Nishine et al. (2012). They used double-lumen catheter to measure lateral airway pressure during bronchoscopy. They have concluded that measurement of the lateral pressure before and after obstruction could estimate the need for additional interventions, better than bronchoscopy alone. They also have observed that the angle of their P-P curve is close to 45 after intervention and close to 0 when airway lumen cross section is reduced.

Similar flow behavior is observed in the present investigation. These studies and others show a new assessment modality by using pressure measurements for the success of interventional bronchoscopy and stent placement, which provides a significant improvement in patient outcomes. Future more, in silico CFD-based pressure prediction, could be used as a valuable tool to provide non-invasive pressure measurements for virtual intervention approaches.

Although this work contributes to the evaluation of the EDAC and functional response to stent placement, there are uncertainties in CFD simulations of biological systems. The assumption of a rigid wall instead of a compliant wall provides reasonable results in normal subjects; however, in patients with EDAC, the trachea wall significantly deforms during exhalation and thus needs to be taken into consideration. In this study, CT-based models were based on the end-point inspiration-expiration, providing extreme geometric models that are used for CFD evaluation of this disease. It is acknowledged that morphing of the geometry between inspiration-expiration is a realistic behavior during respiration; however, the end-point rigid inspiration and rigid expiration CFD simulations provide the maximum and minimum representation of respiration cycle, which are the objectives of this pre- and post-intervention study.

The flow resistance is primarily imposed by conducting airways between 1 and 6 generations of the airways; however, deeper generations of the lung can affect the flow resistance and pressure distributions, even though patient specific CT-images were used to match flow rates and pressures at the outlet boundary conditions. These are currently the subjects of authors' investigations.

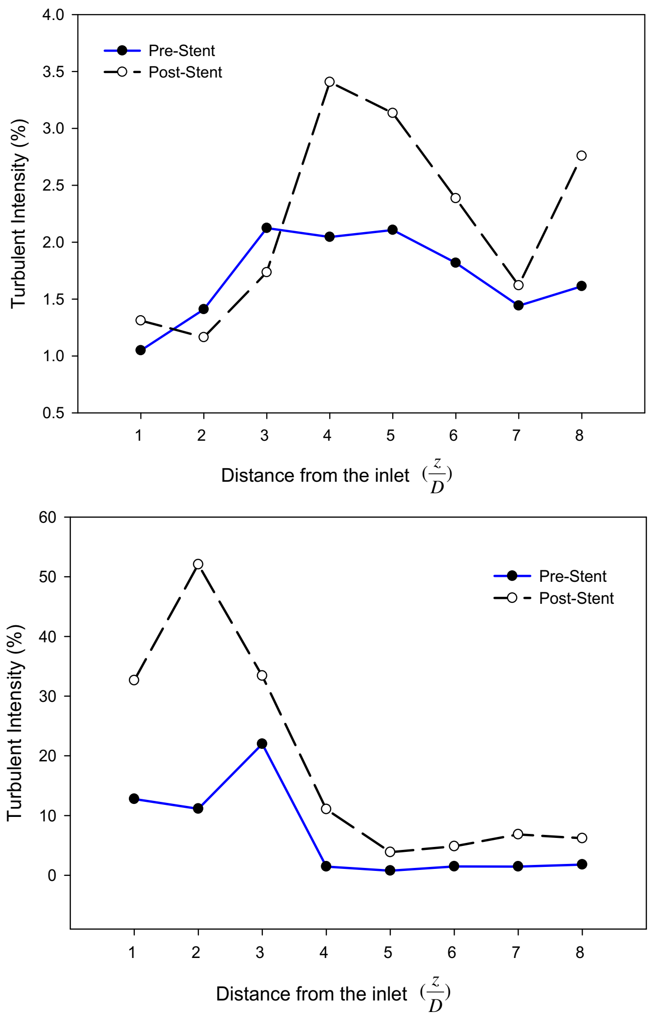

Fig. 8. Turbulent intensity variations pre- and post-stent at peak flow during inspiration (top) and expiration (bottom). The blue solid lines indicate pre-intervention and dashed black lines indicate post-interventions. z is the distance of axial planes in the trachea from the trachea opening downstream, divided by the diameter (D) of the inlet. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

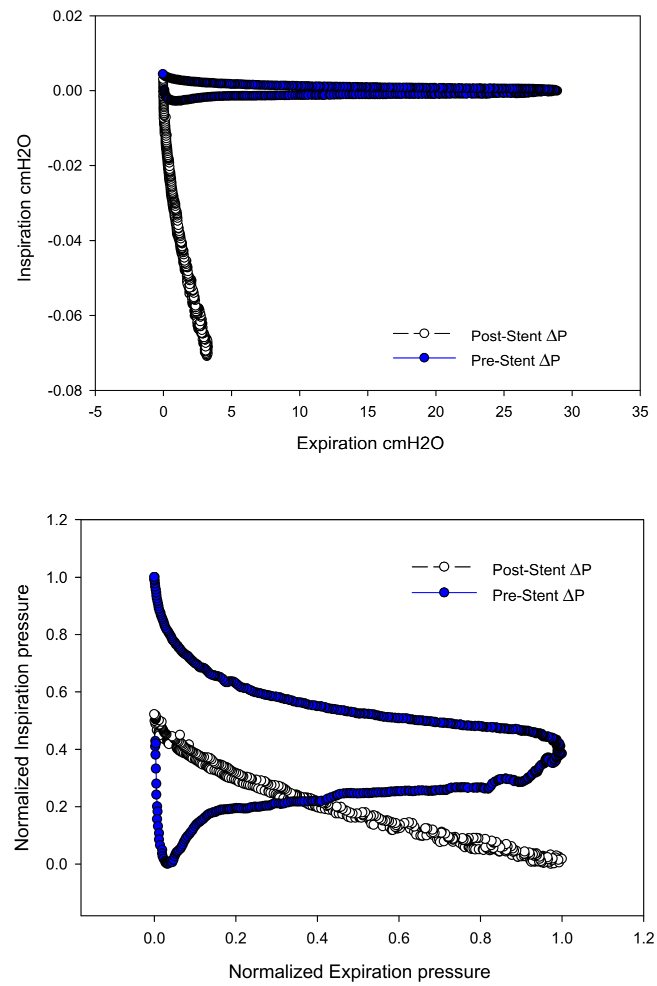

Fig. 9. The pressure-pressure (P-P) curves during quiet breathing; when expressed in cmH_2O (top) and when normalize against the maximum local differential pressure (bottom). ΔP is the pressure drop due to EDAC, measured immediately upstream and downstream of obstructing region.

6. Conclusion

Different diagnosis modalities can be used to diagnose EDAC, which can be categorized into invasive and non-invasive diagnosis. As an intervention modality, stents can provide support for the collapsing airways. Here, CFD simulations of a patient's airways with EDAC, pre- and post-interventions were performed, and the results were compared. Inspiratory-expiratory phase CT-based models were generated and imported into the CFD software for the pre- and post-intervention simulations. The unsteady RANS with k-ω turbulence model, along with patient-specific boundary conditions, were used in the simulations. Results showed that during the inspiratory phase, stenting increases the pressure drop and mean velocity variations; however, the effect is negligible compared to the drastic improvement during the expiratory phase. After the intervention, the WSS averaged in the trachea was also decreased significantly during the expiration, which could suggest improvement in breathing condition. Although the PFT values did not show significant changes for pre- and post-stenting, the CFD results show significant improvement in airflow movement. CFD simulations using CT-based models can add significant values and functional details for evaluation of EDAC disease.

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